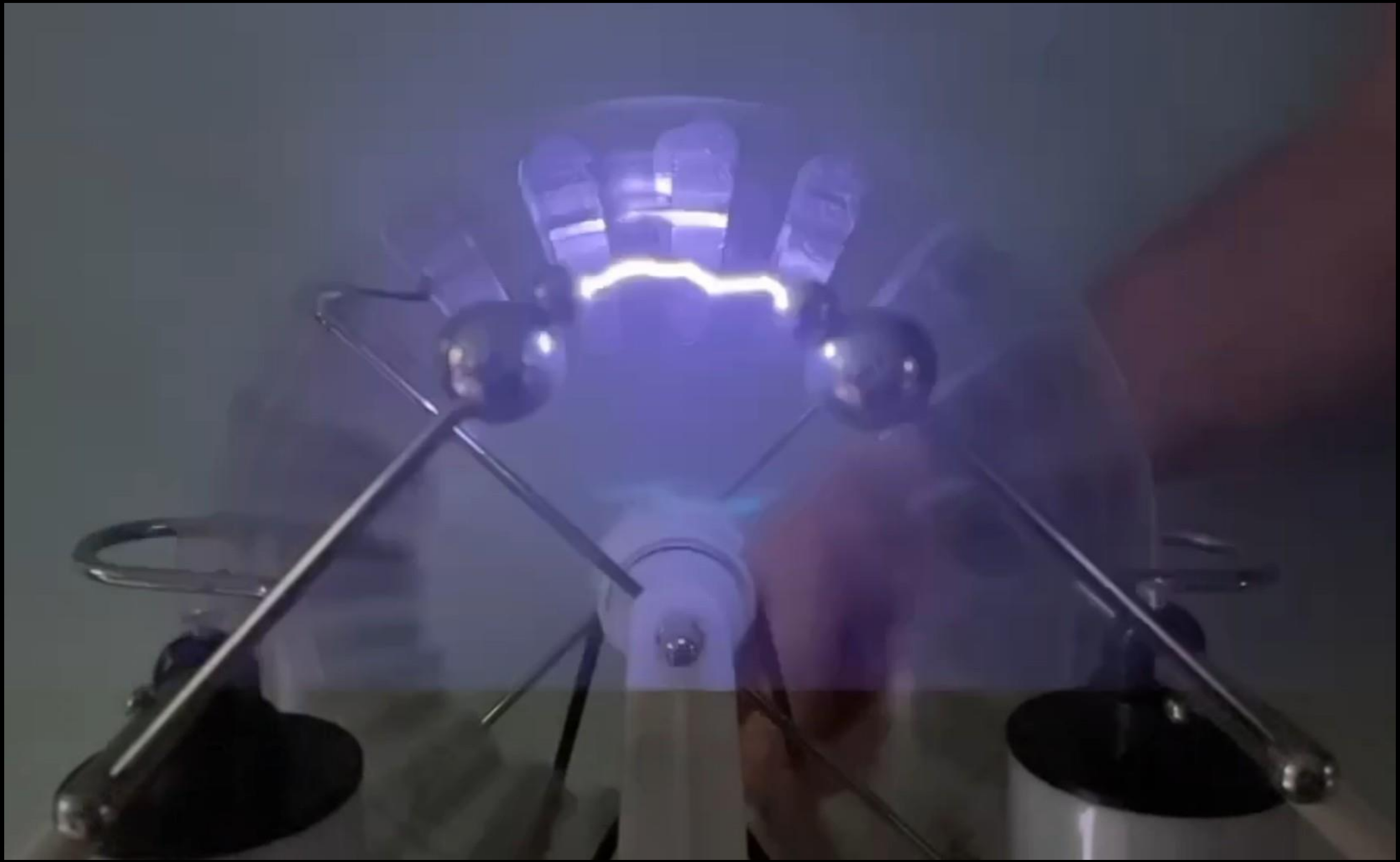


WIMSHURST ELECTROSTATIC GENERATOR

Investigatory Project: From Electrostatic Force to High Voltage, Power & Electromagnetic Emissions

Presentation follows the 60-entry research logbook progression

The Investigation Follows the Logbook

Each phase answers a question raised by the phase before it.

1 Concept & Background	2 Electrostatic Theory	3 Styrofoam Force Demo
4 Experimental Planning	5 Spark & Leyden Jars	6 Voltage, Current & Power
7 EM Emission	8 Applications & Testatika	Final Analysis & Presentation

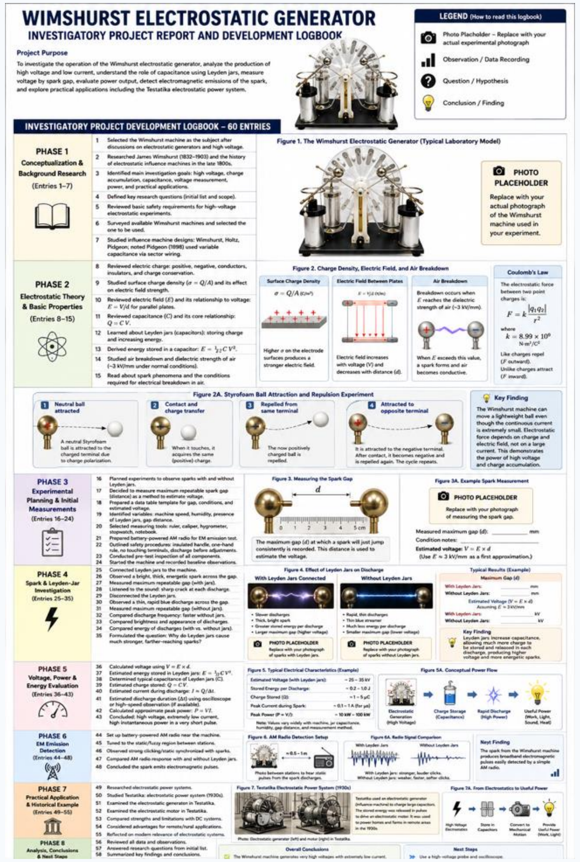
PHASE 1

Conceptualization & Background Research

Entries 1–8

Guiding Question

What does a Wimshurst machine do, and what can we investigate with it?



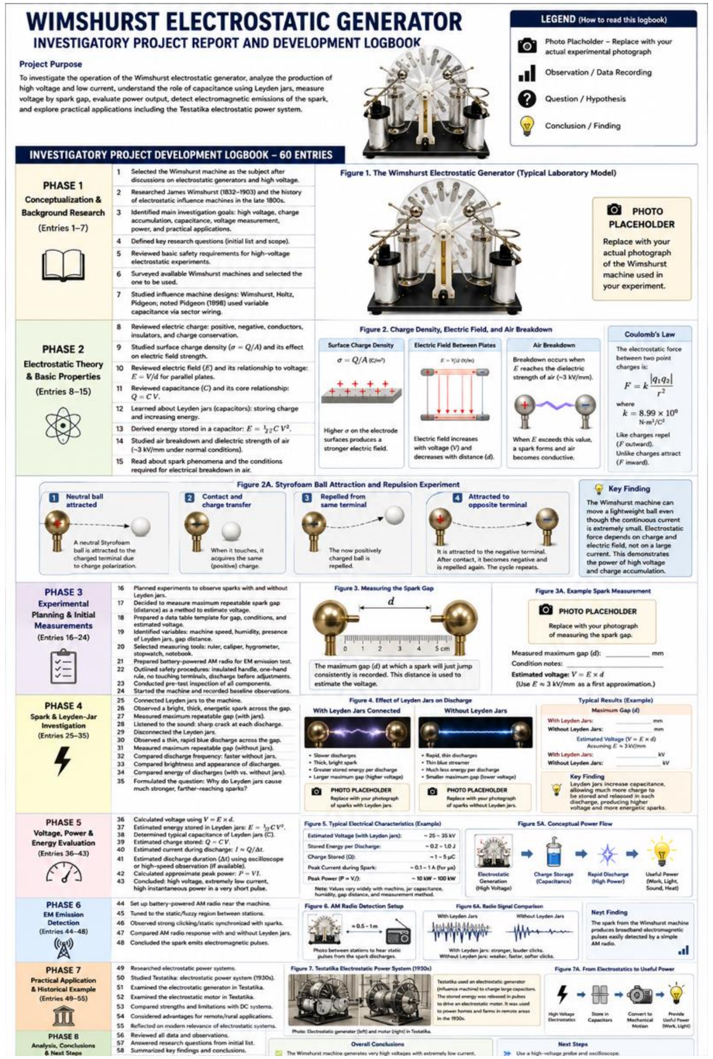
What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

What Does a Wimshurst Machine Do?

Mechanical motion separates charge and creates a very large potential difference.

- Two insulating disks rotate in opposite directions.
- Metal sectors, neutralizer brushes and collectors reinforce charge separation.
- Opposite charges accumulate at the output terminals.
- A sufficiently strong electric field breaks down the air and produces a spark.



Mechanical energy → charge separation → high voltage → spark

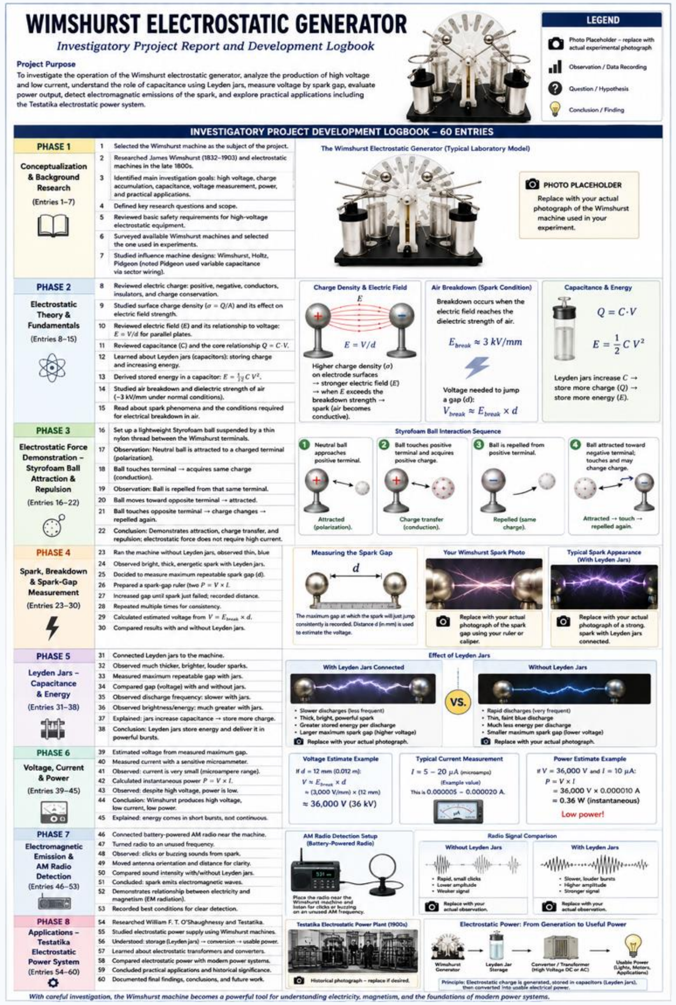
PHASE 2

Electrostatic Theory Development

Entries 9–17

Guiding Question

Which quantities explain charge buildup, electric field, capacitance and air breakdown?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

PHASE 2

Four Quantities to Keep Separate

High voltage, charge, capacitance and current describe different things.

Charge (Q)

How much electrical charge has accumulated.

Voltage (V)

Electrical potential difference between the terminals.

Capacitance (C)

How much charge can be stored for a given voltage.

Current (I)

Rate at which charge flows.

$$Q = C\,V \qquad V = Q / C \qquad \sigma = Q / A$$

Key idea

Low effective capacitance means only a small amount of charge is needed to reach a very high voltage. Surface charge density helps determine the electric field at the electrodes.

PHASE 2

Air Breakdown and Stored Energy

The spark appears when the electric field becomes strong enough to ionize air.

$$E \approx V / d$$

$$\text{Air breakdown} \approx 3\ \text{kV/mm}$$

$$E_{stored} = \tfrac{1}{2} C V^2$$

Electric field

Higher voltage and smaller separation generally increase electric-field strength. Electrode curvature also concentrates the field.

Air becomes conductive

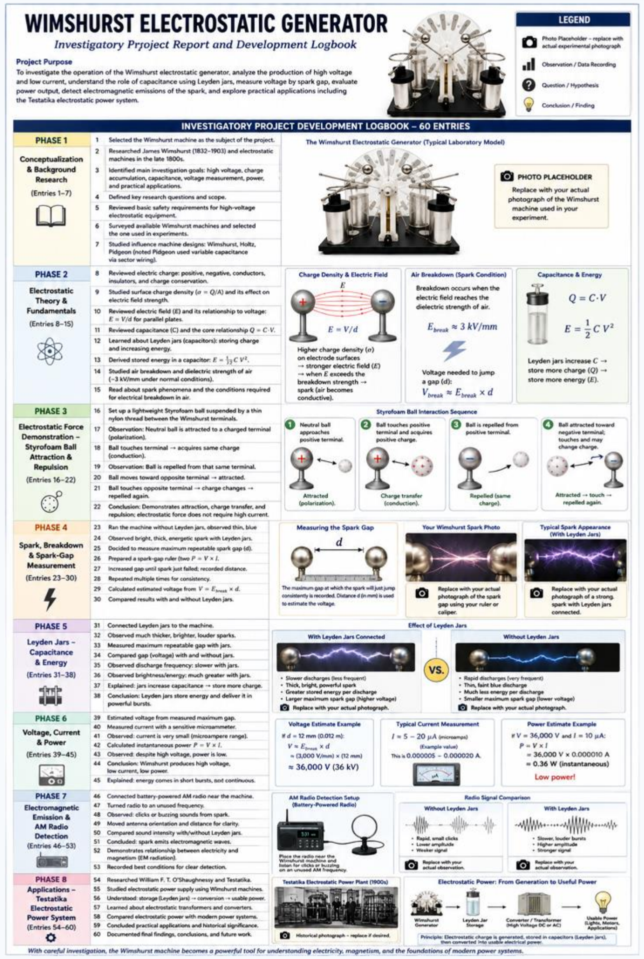
When the field reaches the local dielectric strength of air, ionization creates a conductive path and a spark can form.

Leyden jars store energy

Increasing capacitance lets more charge and more energy accumulate before a discharge.

PHASE 3
Electrostatic Force Demonstration
Entries 18–23

Guiding Question
Can the electric field visibly move matter even though the machine's continuous current is very small?



What this phase adds
Research and observations are organized so each phase builds the explanation used in the final demonstration.

PHASE 3

Styrofoam Ball: Attraction → Charge Transfer → Repulsion

A lightweight suspended ball makes electrostatic force visible.

1

Neutral ball is attracted

Polarization makes the nearer side effectively opposite in charge.

2

Ball touches terminal

Charge transfers by conduction.

3

Ball is repelled

After contact it has the same polarity as that terminal.

4

Ball moves to opposite terminal

It is attracted, can touch, change charge, and be repelled again.

$$F = k\,|q_1 q_2| \,/\, r^2 \qquad k \approx 8.99 \times 10^9\ \mathrm{N{\cdot}m^2/C^2}$$

PHASE 3

What the Styrofoam Experiment Demonstrates

Electrostatic force depends on charge and separation—not on a large continuous current.

Why a neutral ball is attracted

The charged terminal polarizes the neutral ball. Opposite charge is effectively closer to the terminal than like charge, producing a net attractive force.

What changes after contact

Touch transfers charge. The ball can acquire the same polarity as the terminal and is then repelled.

Why this matters

The Wimshurst machine can produce a strong enough electric field to move a very light object even though its available continuous current is extremely small.

Unlike charges attract • Like charges repel

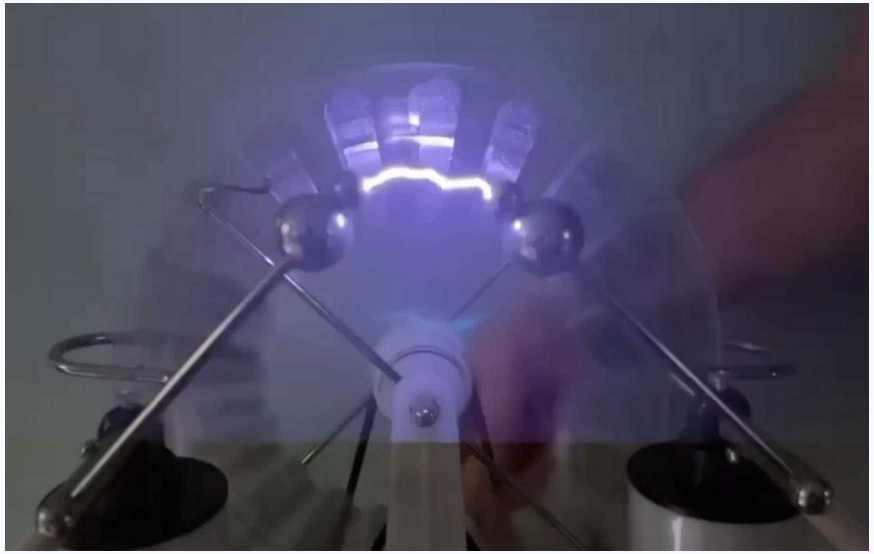
PHASE 4

Experimental Planning & Measurement Design

Entries 24–27

Guiding Question

How can we compare the two discharge conditions fairly and estimate the high voltage safely?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

PHASE 4

Plan the Spark-Gap Experiment

Use the maximum repeatable spark gap as the primary quantitative measurement.

Measure

Use a ruler or caliper to measure electrode spacing. Repeat the measurement rather than relying on one spark.

Control

Keep electrode geometry, approximate cranking rate and room conditions as consistent as practical. Compare jars connected vs. disconnected.

Record

Gap distance • discharge frequency • brightness • thickness • sound • room conditions • Leyden-jar condition.

$$\text{Voltage estimate:} \quad V \approx (3\ \text{kV/mm}) \times d$$

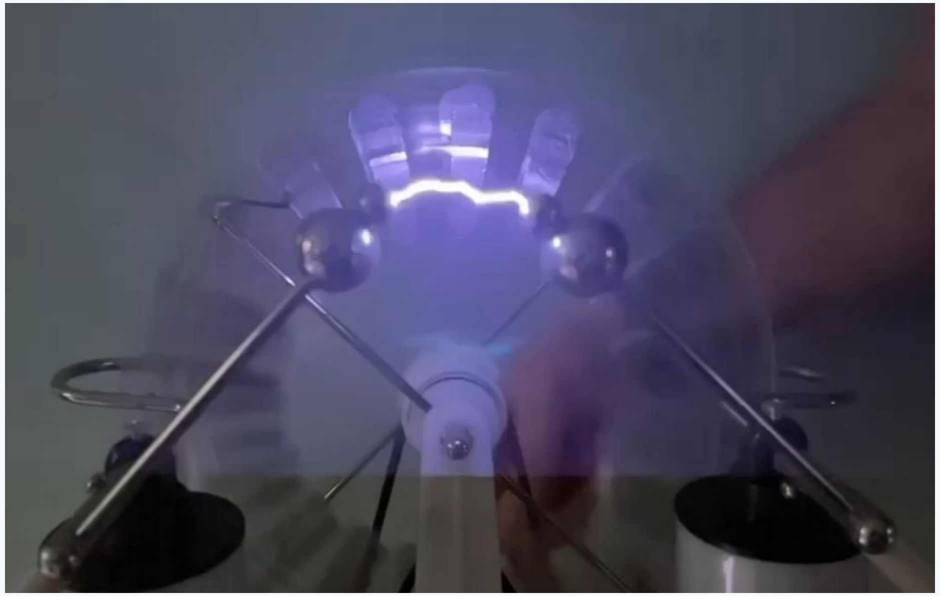
PHASE 5

Spark & Leyden-Jar Investigation

Entries 28–36

Guiding Question

Why does removing the Leyden jars create a rapid thin blue discharge instead of large bright sparks?

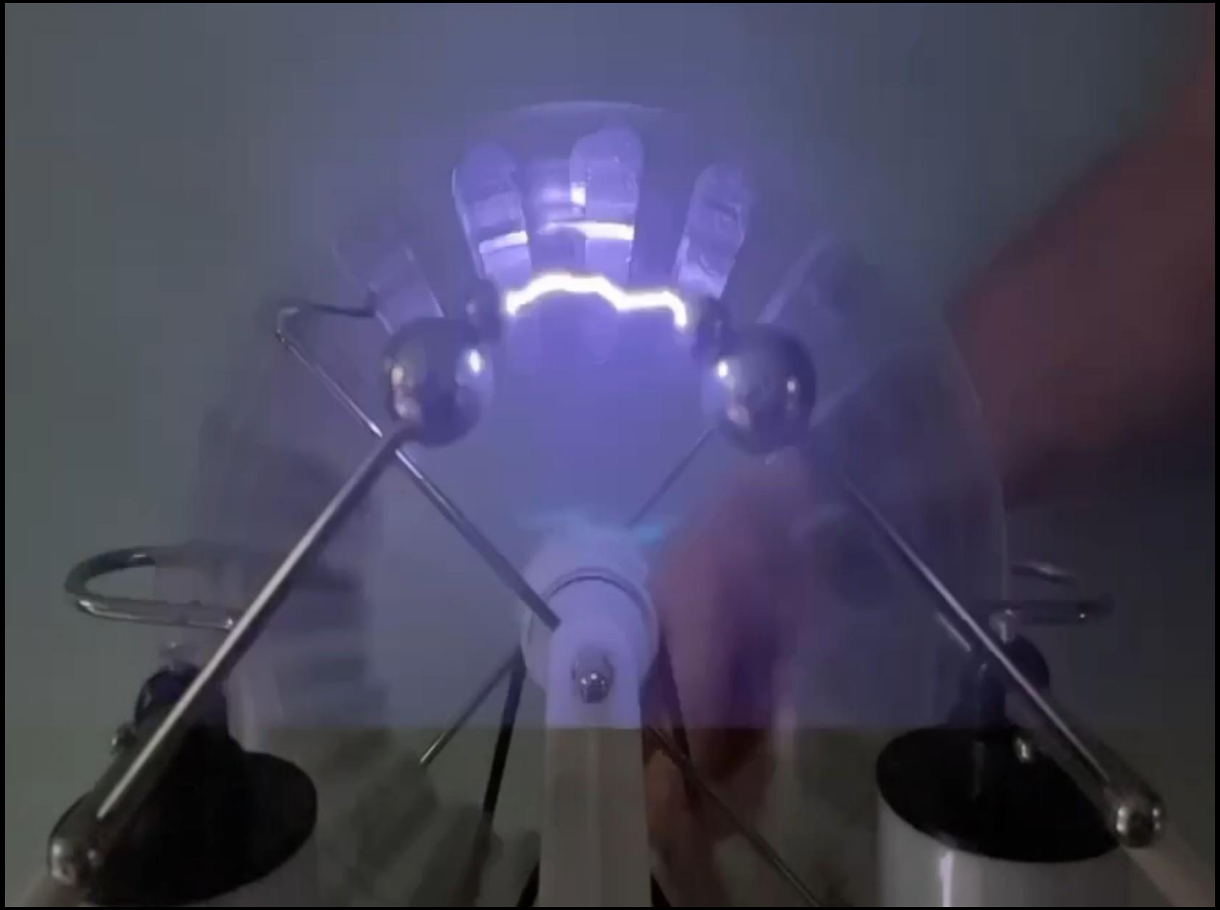


What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

PHASE 5

With Leyden Jars: Slower, Brighter, More Energetic Sparks



What changes?

- The jars greatly increase capacitance.
- More charge is needed to reach the breakdown region.
- Charging takes longer.
- Much more charge and energy are available when breakdown occurs.
- The result is a thick, bright, loud spark.

$$E = \tfrac{1}{2} C V^2$$

PHASE 5

Without Leyden Jars: Rapid Thin Blue Discharge

Lower effective capacitance changes the discharge regime.

What is happening?

- Effective capacitance is small.
- A little transferred charge produces a large voltage rise.
- Surface charge creates a strong electric field.
- Air ionizes quickly.
- The small stored charge discharges, then the cycle repeats rapidly.

Key observation

The discharge can look like a rapid, thin blue stream rather than isolated bright sparks. Each event contains much less stored energy, but the machine reaches breakdown again very quickly.

Small C → little Q needed → breakdown reached quickly

PHASE 5

Same Generator — Very Different Discharge

The Leyden jars primarily change charge storage and discharge energy.

WITHOUT LEYDEN JARS

Low capacitance

Voltage rises quickly

Frequent discharge

Thin blue stream

Little energy per event

WITH LEYDEN JARS

Higher capacitance

More charge must accumulate

Less frequent discharge

Thick, bright spark

Much more energy per event

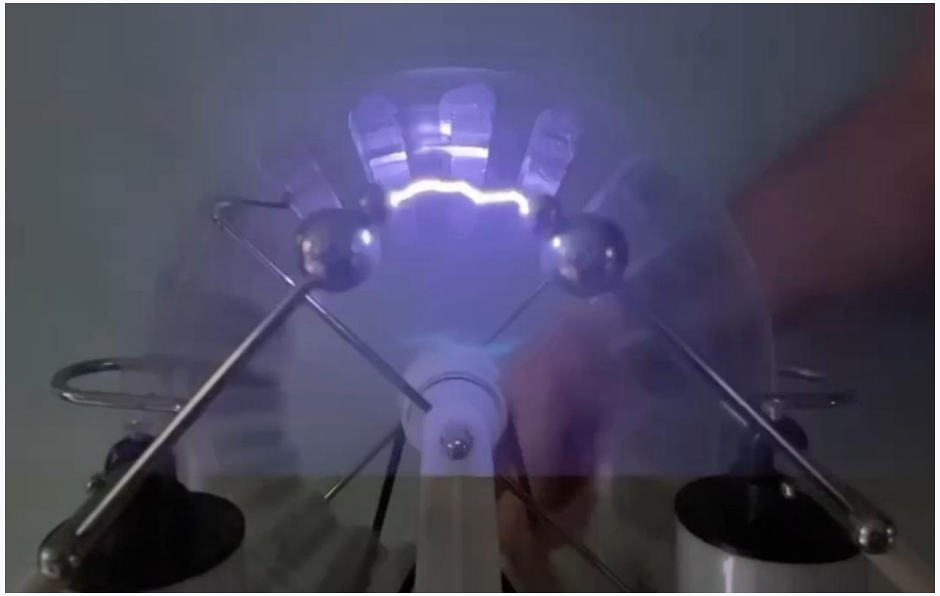
PHASE 6

Voltage, Current & Power Analysis

Entries 37–44

Guiding Question

How high is the voltage, why not measure it with an ordinary meter, and how much power does high voltage actually represent?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

PHASE 6

Estimating Voltage with the Spark Gap

Air breakdown provides a practical indirect estimate.

$$V \approx (3\ \mathrm{kV/mm}) \times d$$

Example

If the maximum repeatable gap is 10 mm:

10 mm × 3 kV/mm ≈ 30 kV

Why it is only an estimate

Air breakdown varies with humidity, pressure, temperature, electrode shape, surface condition and other factors. Use several trials and report the maximum repeatable gap.

PHASE 6

Why Not Use an Ordinary Voltmeter or Oscilloscope?

Direct measurement requires specialized high-voltage, very-high-impedance instrumentation.

Voltage rating

A Wimshurst machine may reach tens of kilovolts. Ordinary meters and probes are generally not rated for direct connection at these voltages.

Loading

A normal meter has finite input resistance. On a very-low-current electrostatic source, the meter can draw enough current to collapse the voltage instead of simply measuring it.

Oscilloscope ground

Many oscilloscopes use an earth-referenced probe ground. Connecting that ground to a floating electrostatic generator can alter the circuit and create an unintended discharge path.

PHASE 6

High Voltage Does Not Necessarily Mean High Power

Power depends on both voltage and current.

$$P = V I$$

Illustrative Wimshurst calculation

$V = 30{,}000$ V
$I = 10\ \mu A = 0.000010$ A

$P = (30{,}000)(0.000010) = 0.30$ W

Compare: USB supply

$5\ V \times 2\ A = 10\ W$

The USB supply has 6,000× less voltage yet can deliver more than 33× as much continuous power in this example.

30,000 volts sounds enormous — but at 10 millionths of an ampere, the calculated continuous power is only 0.30 watt.

PHASE 6

Why Can a Low-Power Machine Make Such a Dramatic Spark?

Stored energy can be released in a very short time.

$$E = \tfrac{1}{2} C V^2$$

$$P_{discharge} \approx E / t$$

Energy accumulation

A Leyden jar may charge over a relatively long interval, then release much of its stored energy in a tiny fraction of a second.

Example only

0.05 J ÷ 0.0001 s = 500 W during that brief interval.

That does NOT mean the generator continuously produces 500 W.

PHASE 6

Supplementary Demonstration: Lighting a Neon Lamp

High voltage can ionize neon and produce visible light with very small current.

What it demonstrates

A small neon lamp can glow or flash because the Wimshurst voltage is high enough to ionize the gas inside the lamp. This provides a second visible indication of high voltage besides the spark.

What it does NOT demonstrate

A glowing neon lamp does not imply large current or large continuous power. It reinforces the distinction between voltage, current and power.

High voltage ≠ high current ≠ high power

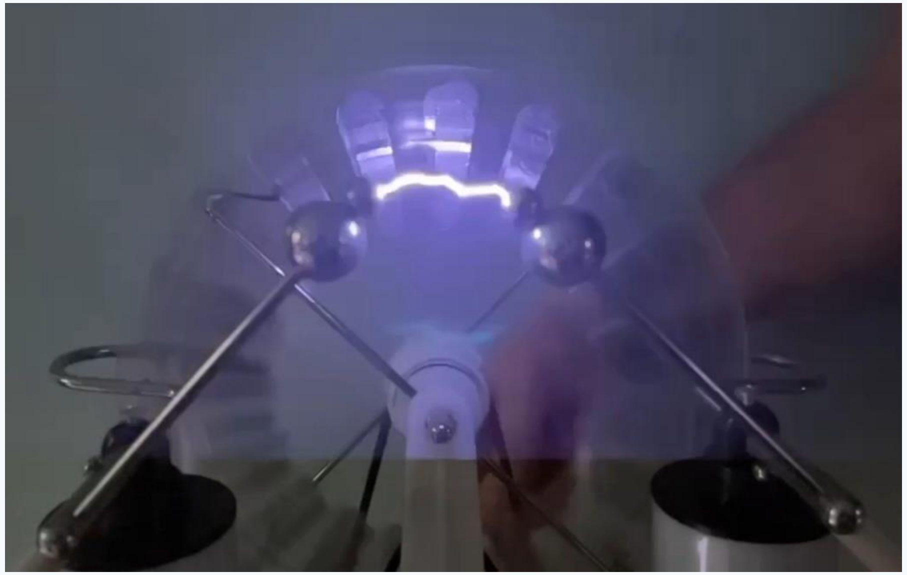
PHASE 7

Electromagnetic Emission Investigation

Entries 45–50

Guiding Question

Does the spark also send out an electromagnetic signal that can be detected without a wire connection?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

PHASE 7

A Spark Is Also an Electromagnetic Signal

A rapid electrical discharge produces a broadband electromagnetic disturbance.

Spark → fast-changing current/field → electromagnetic pulse → AM radio click

Historical connection

Early spark-gap transmitters intentionally used electrical sparks to generate radio-frequency energy.

Why compare the jars?

The two discharge regimes differ in repetition rate and energy per event, so the radio should provide an audible comparison.

PHASE 7

Experiment: Detect the Spark with an AM Radio

Compare the radio interference with and without the Leyden jars.

Setup

1. Battery-powered AM radio
2. Tune between stations
3. Start about 1–2 m away
4. Keep volume fixed
5. Use the same spark gap and similar cranking speed
6. Optionally compare maximum clear detection distance

Without Leyden Jars

Expected: rapid, weaker impulses may sound like fast crackling or buzzing.

The discharge repeats quickly because little charge is stored per event.

With Leyden Jars

Expected: less frequent but stronger individual clicks or pops.

Each spark releases more stored energy.

PHASE 8

Practical Applications & Testatika Extension

Entries 51–55

Guiding Question

How can electrostatic high voltage be useful, and what additional stages would be needed to supply conventional loads?

What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

PHASE 8

Practical Uses of High Electrostatic Voltage

High voltage itself is useful even when available current is small.

Electrostatic precipitation

Remove particles from air or exhaust streams.

Electrostatic coating

Attract charged paint or powder to a target surface.

Ionization

Create ions for demonstrations, sensing or processing.

Particle acceleration

Use electric fields to accelerate charged particles.

High-voltage research

Study insulation, corona, breakdown and electric fields.

PHASE 8

From Electrostatic Generation to Practical Electrical Power

The central engineering problem is not just producing voltage—it is converting and conditioning energy for a load.

Electrostatic source
Very high voltage
High source impedance

→

Charge storage
Capacitance
Energy buffering

→

Conversion
Pulsed / resonant inductive stages

→

Conditioning
Rectification regulation / inversion

→

Useful load
Light, motor, electronics

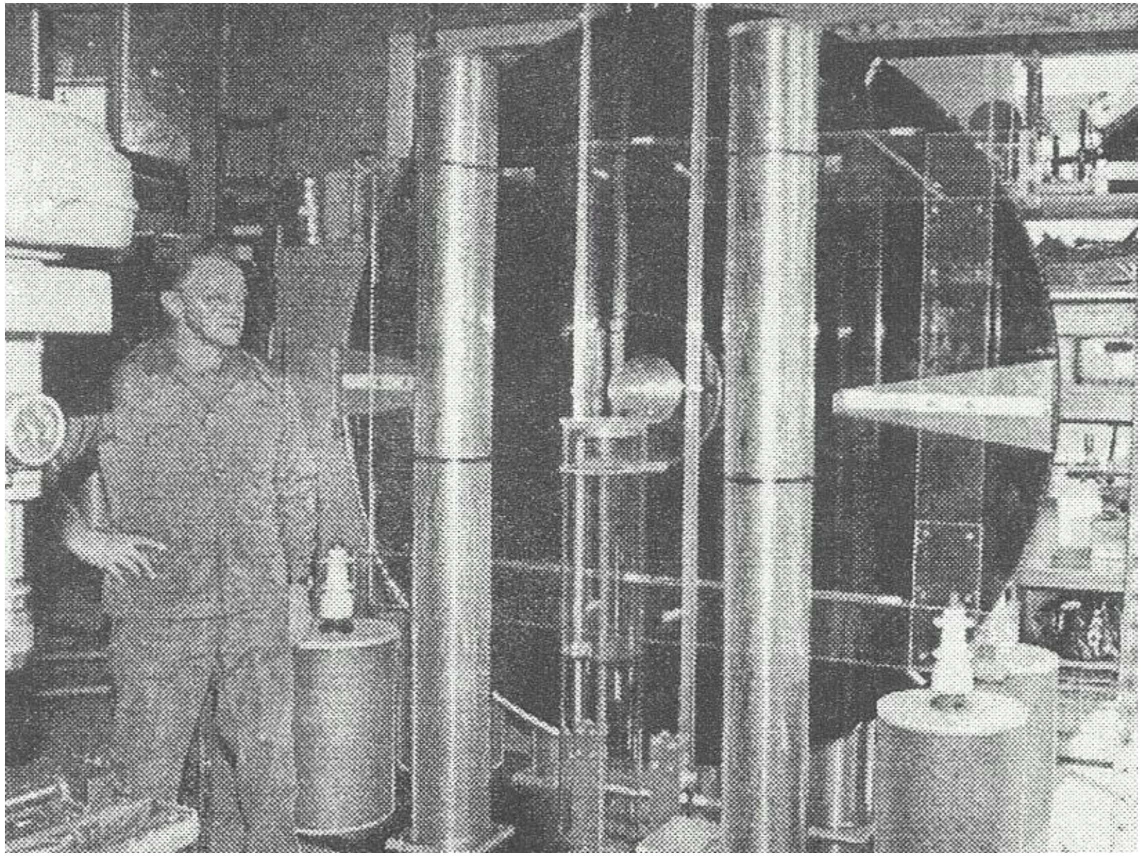
Inductive loads

Motors and transformers may require starting current, reactive current and transient handling. A practical conversion system must isolate the electrostatic source from those load demands.

PHASE 8

Testatika: A Reported Extension of the Influence-Machine Idea

For this project, it is presented under the premise that its reported operation is valid.



Conceptual relationship

The electrostatic disk system can be viewed as a high-voltage front end, followed by additional storage, inductive/resonant conversion and output-conditioning structures.

Important distinction

Photographs establish that the Testatika contains much more than rotating disks. Exact internal wiring and the function of every component are not fully documented in public primary engineering drawings, so detailed circuit descriptions should be labeled as reported or reconstructed.

Final Phase — Analysis, Documentation & Presentation

Entries 56–60: assemble the observations into one coherent scientific explanation.

Final Demonstration Sequence

1. Styrofoam attraction / repulsion
2. Spark gap
3. Compare jars vs. no jars
4. Estimate voltage
5. Explain voltage, current & power
6. Neon lamp demonstration
7. AM-radio detection
8. Practical applications / Testatika

What the project establishes

- Electrostatic force can act without large current.
- Capacitance changes charge and energy storage.
- Air breakdown provides an indirect voltage estimate.
- High voltage can coexist with low continuous power.
- Sparks produce detectable electromagnetic emissions.

What to record

Replace AI placeholders with actual photos where possible. Record measured spark gaps, room conditions, qualitative discharge observations and AM-radio results. Distinguish measurements from illustrative examples.

Overall Conclusion

The coherent story

Mechanical motion separates charge. Charge creates electric fields and electrostatic force. Capacitance determines how much charge and energy can accumulate. Air breaks down when the electric field becomes strong enough. The resulting spark can be used to estimate voltage and can radiate a detectable electromagnetic pulse. Yet high voltage alone does not imply high continuous power. Practical power delivery requires storage, conversion, conditioning and load matching.

Electrostatic force → high voltage → capacitance → breakdown → power → electromagnetism → applications

Safety

High voltage requires respect.

- Do not deliberately discharge the machine through a person.
- Treat Leyden jars as charged capacitors even after cranking stops.
- Discharge them with an appropriate insulated discharge tool before touching conductors or changing connections.
- Keep away from flammable vapors and sensitive electronics.
- Do not connect ordinary meters or oscilloscope probes directly to the generator.
- Keep people with implanted electronic medical devices away from the discharge area.
- Perform the demonstration with responsible adult supervision.

Perspective

High-voltage sources capable of sustained current can be far more dangerous than a Wimshurst machine.

Project Source Materials

The deck reorganizes the existing presentation content according to the authoritative logbook flow.

Primary flow

Wimshurst_60_Entry_Logbook_A4_With_Styrofoam_Experiment.docx — phase sequence, entries, Styrofoam experiment, formulas and final demonstration progression.

Existing presentation content

Wimshurst_Electrostatic_Generator_Presentation_Expanded.pptx — voltage/current/power explanation, spark-gap measurement, meter/scope limitations, Leyden-jar comparison, neon lamp, AM-radio experiment, safety and Testatika extension.

Supporting Testatika material

Testatika_History_Architecture_and_Power_Conversion_Report — used only to preserve the distinction between visible/reported architecture and later reconstruction hypotheses.

Holtz_Wimshurst_Pidgeon_Testatika_Student_Research_Guide — supplemental research guide examining the evolution from Holtz to Wimshurst to Pidgeon and ultimately the Testatika, emphasizing electrostatic induction, variable capacitance, charge storage, and the development of electrostatic energy-conversion concepts.

Research Extension: Converting Wimshurst Output into Usable Power

Treat the machine as a high-voltage charge source feeding an energy-conversion chain.

Wimshurst → HV storage → controlled pulse → LC / transformer → rectification → output storage → load

SOURCE	STORAGE	CONVERSION	CONDITIONING	LOAD
High voltage Tiny current	Capacitance $E = \frac{1}{2}CV^2$	Pulse / LC Transformer	Rectify buffer / regulate	Low voltage ≈ 0.25 W

The converter does not create energy.
It transforms a very-high-voltage/high-impedance source into a lower-voltage/higher-current form, with losses at each stage.

Could Approximately 0.25 W Be Achieved?

The answer depends on measured average charging power.

Illustrative source

$$25{,}000\ \text{V} \times 20\ \mu\text{A}$$
$$= 25{,}000 \times 0.000020$$
$$= 0.50\ \text{W}$$

Illustrative conversion

$$90\% \times 70\% \times 85\% \times 90\%$$
$$\approx 48\%$$
$$0.50\ \text{W} \times 0.48 \approx 0.24\ \text{W}$$

These values are examples—not measurements of this Wimshurst machine.

Measure a known capacitance and charging time: $E = ½CV^2$ and $P_{avg} \approx E/t$.
This determines whether the 0.25 W target is realistic for the actual apparatus.

Proof-of-Concept Development Sequence

Build and measure one energy-conversion stage at a time.

1. Characterize charging of a known, safely rated capacitance.
2. Calculate stored energy $E = ½CV^2$ and average charging power E/t.
3. Transfer stored energy into an enclosed inductive / resonant stage.
4. Step down the pulse with a transformer or LC/transformer network.
5. Rectify and buffer the lower-voltage output.
6. Apply a calibrated resistive load and measure Vout and Iout.
7. Calculate Pout = VoutIout and conversion efficiency.
8. Demonstrate a small LED/load after the measurements are understood.

Safety: added high-voltage capacitance can store dangerous energy. Keep the HV section enclosed, intentionally discharged, and adult-supervised.